INSTRUCTION FOR PACKING, MARKING AND SHIPPING

CONTRACT NO. : **POGC-760-81-156**

PROJECT : **SOUTH PARS FIELD DEVELOPMENT (PHASES 9&10)**

COMPANY : **NATIONAL IRANIAN OIL COMPANY**

SITE : **Pars Special Energy & Economic zone (ASSALUYEH) / PERSIAN GULF, IRAN SOUTH PARS GAS FIELD**

A	03/06/2003	Issued for Company	LG	HJY	MSO		SHH
REV.	**DATE**	**DESCRIPTION**	**ORIG.**	**PRPD**	**CHKD**	**APP'D**	**CONTRACTOR's APP'D**

TABLE OF CONTENTS

1. PURPOSE

2. TERMS AND DEFINITIONS

3. SCOPE

4. GENERAL

5. RESPONSIBILITIES AND WARRANTIES

6. PREPARATION FOR PACKING

7. HEAVY LIFT OUT OF GAUGE EQUIPMENT

8. CONTAINERIZATION

9. MARKING INSTRUCTIONS AND STORAGE CODE

10. PACKING LIST

11. PACKING INSPECTION – PACKING PERFORMANCE

12. SHIPPING INSTRUCTION

ATTACHMENT NO.1 PURCHASER'S Logistics Organization
NO.2 Freight Forwarder's Organization
NO.3 Notice of Vendor's Responsible Personnel for Transportation
NO.4 Safety Data Sheet
NO.5 Preliminary Shipping Information for Space Booking
NO.6 Commercial Invoice
NO.7 Certificate of Origin

ANNEX I MAIN SHIPPING MARK

ANNEX II EQUIPMENT STORAGE CODE

ANNEX III MARKING SPECIFICATION

ANNEX IV PACKING LIST

ANNEX V PACKING FOR PIPING MATERIAL

ANNEX VI PACKING FOR STRUCTURAL STEEL

ANNEX VII EUROPIAN INDUSTRIAL PACKING SPECIFICATION (SEI) - 1992

1. PURPOSE

This document applies for using as an instruction for PACKING, MARKING AND SHIPPING in accordance with CONTRACT, POGC-760-81-156.

2. TERMS AND DEFINITIONS

- Supplier : The company to whom a purchaser order or contract has been placed by the contractor.
- Contractor : LG E&C / O.I.E.C. / I.O.E.C. Consortium

3. SCOPE

The supplier shall prepare pack and mark all supplies in accordance with the present INSTRUCTION for PACKING, MARKING AND SHIPPING.

4. GENERAL

4.1 Packing means to efficiently and effectively protect goods, in supplier's time schedule, for a period up to 12 months from EXW delivery. Packing shall be sufficient to withstand, without irritation, rough handling during loading/unloading, sea and/or inland transport, bad handling during transit, and outdoor storage. Depending on their type, supplies are to be protected against both mechanical damage(shocks, rupture, breakage, loss or pilfering) and corrosion(rain, salty atmosphere, etc.,)

4.2 Equipment and materials must be export packed as per European Industrial Packaging Specification such as SEI 1992 EDITION. Categories of packing are defined in the last 1992 Edition of SEI.

4.3 Packing procedure and packing list shall be submitted to contractor for approval, and have to get approval prior to commencing the packing work and notifying the packing inspection application.

4.4 Each Korean vendor shall assign only the special packing companies which are pre-qualified by LG E&C.

4.5 Packing list, packing note and marking of the packages shall be in English.

5. RESPONSIBILITIES AND WARRANTIES

5.1 Supplier shall be responsible for compliance with these instructions and the choice of category for packing, according to the nature of the supplies and transport facilities used.

5.2 Contractor reserves the right to inspect the packing at any time, and to reject it when it does not meet packing requirements.

5.3 Supplier shall be responsible for the accuracy of the weights and dimensions declared, and marking of packages. The packing list shall be in strict conformity with the packing content.

5.4 All costs incurred in connection with missing or damaged supplies caused by non-compliance with these instructions shall be charged to the supplier.

5.5 After notification by contractor, supplier shall take immediate action for the replacements of such missing or damaged supplies within an acceptable time schedule, at supplier's own cost.

5.6 Supplier shall be responsible for recommendations for storage of the supplies for 12 months according to site conditions. Any damage incurred due to faulty or missing recommendations shall be borne by the supplier.

5.7 Supplier shall be responsible for any loss or damage caused through wrong marking.

6. PREPARATION FOR PACKING

Prior to packing, equipment shall be protected under supplier's responsibility as hereunder described and not limited to :

6.1 GENERAL

6.1.1 The goods shall be dried, cleaned internally and externally, removed from all welded spatter, scale, rust, cuttings, fittings, etc., as well as any all foreign material.

6.1.2 Protection against environmental corrosion caused by extremes of humidity, condensation, rain, dust, sand, salted air and sea spray shall be provided. If the order stipulate the high alloy steel (stainless steel etc.) shell is to be treated with acid cleaning, the equipment must be protected by vinyl wrapping after cleaning.

6.1.3 Critical wearing surfaces, highly stressed parts, and highly finished surfaces shall be further cleaned by dipping or brushing with a suitable product.

6.1.4 The goods being delivered by partial unit or partially dismantled shall be furnished with suitable markings required for assembly and installation at the Site.

6.1.5 Unless otherwise specified, after hydrotesting, operation, and/or performance test, all fluids, i.e, lubricating oil, fuel oil water, etc., shall be completely drained from all tanks, vessels, jacketing, piping, etc., and wiped or blown dry.

6.1.6 When the packing is carried out in boxes and/or crates, the metal or plastic corners at the

slinging points shall be used.

6.1.7 The piping material will be packed as per Annex V.

6.1.8 Specific points due to the requirements of the destination, terms of the contract etc., will be specified separately.

6.1.9 If the order stipulates that the shell is be treated with anti-corrosive and/or painted, the equipment, the equipment must be handled with particular care. If the order stipulates that the stainless steel is to be treated with acid cleaning, the equipment must be protected by vinyl wrapping after cleaning.

6.2 SMALL PARTS, TOOLS AND SPARES PARTS

6.2.1 Small loose parts
Small loose parts shall be tagged and packed in a separate, sturdy wooden box. Any such box shall be shipped with main equipment, being securely attached to the main skid or within the main box. Secure attachment consists of steel strapping or bolting. Tiring or wiring are prohibited.

6.2.2 Special tools
Tools shall be packed in an individual, rugged wooden box and marked as “special tools”.

6.2.3 Spare parts for commissioning, start-up and for 2 years operation.

Each category of spare parts shall respectively be packed in an individual rugged wooden box, packed separately and marked :
- commissioning, start-up (COMMISSIONING SPARES)
- two years maintenance (OPERATING SPARES)

7. HEAVY LIFT OUT OF GAUGE EQUIPMENT

7.1 DIMENSIONS AND WEIGHT

Shall be classified as OUT OF GAUGE or HEAVY LIFT, equipment which characteristics exceed on of the following criteria :

Length	:	12	m
Width	:	2.5	m
Height	:	2.5	m
Weight	:	20	MT

7.2 INTERNAL SUPPORT

For large size of equipment, such as, columns, reactors, pressure vessels, stack/chimney etc.. Supplier shall provide internal supports to prevent deformation.

7.3 LOAD DISTRIBUTION

As far as transport drawings issued by supplier are confirmed, supplier shall control the adequacy of the supports and/or saddles and ensure that load distribution now fixed at 10

tonnes/m2 (to be confirmed in due time), is in compliance with contractor's nominated Freight Forwarder Requirements with respect to road, rail and sea transportation and unloading port requirements.
In case supplier has the responsibility to deliver on a CPT site basis, supplier shall take into consideration, prior to saddle fabrication the maximum load distribution allowed with respect to local means of transportation and handling and port requirements.
In case of CFR nominated port delivery term basis, supplier shall liaise with contractor's nominated Freight Forwarder with respect to local load distributions tolerance prior to saddle fabrication.

7.4 PREPARATION OF SEA FASTENING OPERATIONS

For any contractual delivery terms stated on the purchase order (EX WORKS, FOB, FCA etc.,), the supplier shall carry out the preparation of sea fastening according to the instructions which will be given by the shipping company through contractor's forwarder and the following.

The unpacked equipment shall be prepared for sea fastening operations and be equipped as follows :

a. Lifting lugs to be calculated, defined and shown on the transport drawing. They will be used for any handling operations onto trucks for road transportation and onto vessels for sea transportation and onto wagon for rail transportation ; wireguard included.
b. Lashing points shall be calculated in order to support all forces crosswise, lengthwise, and so on. They will be clearly marked on the equipment and transport drawings
c. When the specificity of equipment requires utilization of special tools, such as lifting beams, slings, etc., for handling operations, the supplier SHALL PREPARE A LIST OF SPECIAL TOOLS WHICH WILL BE PROPOSED FOR contractor's approval and imperatively included in the Purchase order.
   The supplier shall be responsible for the non-observance of the above.

8. CONTAINERIZATION

Instrument and sophisticated electrical Panels nominated by LG E&C shall be stuffed in Dry Container in order to protect from damage. When packing cargo is made by wooden box or crate, Vendor shall try to make a package less than following dimensions for container carriage.

20FT container : 550 × 220 × 205cm (L × W × H)

40FT container : 1150 × 220 × 205cm (L × W × H)

At supplier's cost, the supplier or their packer shall carry out the stuffing operation of goods into container when required.

9. MARKING INSTRUCTION AND STORAGE CODE

Markings for each package shall be in accordance pattern in ANNEX I.

9.1 TYPE AND SIZE OF GRAPHICS

Packages and case win be marked with indelible black paint resistant to sea water.
Marking must be perfectly legible with the contrasting color.
Marking Specification for Boxes, Crates and Shrink - Wrapped Packages as per indication ANNEX III

Ex. : If the background to be marked is “black” or “dark color”, the marking should be “white”.

Stencil on two opposite sides and the top of each package in clear characters at least 40 mm high (where case size permits, otherwise use optimum size for each package dimension).

N.B. : Metal tags or labels must be stamped or indelibly marked with full shipping marks and must be securely attached, welded or riveted to all loose, bundles or uncrated items.
Cable drums : On the two flanges.

Dimension of case or packages will be indicated in cm and the weight in kg. No other units shall be used.

9.2 STORGE CODE

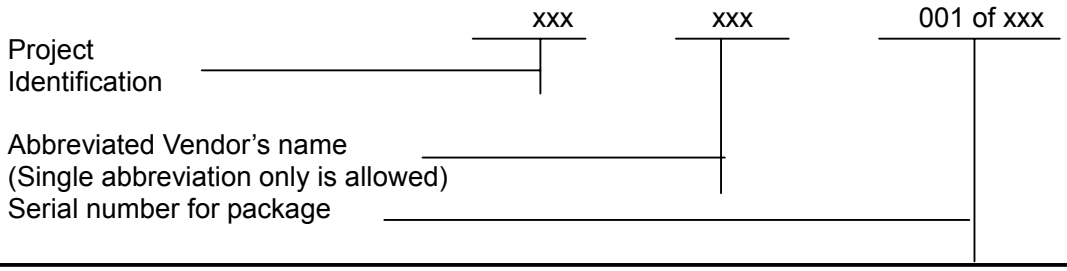
The type of storage code to be printed on the two opposite sides and top of each package is specified as follows :

X - ► Crates or Packages to be stored outside
XX - ► Crates or Packages to be stored under cover / or open shed
XXX - ► Crates or Packages to be stored in covered or closed premises
XXXX - ► Crates or Packages to be stored in air conditioned closed promises

Refer to Equipment Storage code in ANNEX II

9.3 PACKING NUMBER

In principle, the package number within the main shipping mark of each package shall be as follows :



Abbreviated Vendor's name shall be used after confirmation from contractor.
The first number being that of the order in the batch, then after oblique, the last number being the total in the batch. For example, for a batch of 100 package :
First package no. : 001 of 100 & Last package no. : 100 of 100.
Boxes to be included inside package shall be numbered with a sub number on the surface of box.
For example, if the package number of the outside box is xxx-(xxx)-001 of 010, the inside box number shall be xxx-(xxx)-001 of 010-1 and up.

9.4 INTERNATIONAL STANDARDS SYMBOLS

9.4.1 For handling

All boxes and creates shall also be marked with appropriate international standard symbols for handling as shown in the French Industrial Packaging Specification SEI 1992 EDITION.
All boxes and crates are to be marked clearly on all low sides :

"HANDLE WITH CARE" and "RIGHT SIDE UP"

In addition, each box shall be marked "KEEP DRY" on all four sides.

In the case of packages with a single gross weight totaling 2,000kgs and/or height of more than 1 m, the centre of gravity shall be clearly marked with the symbol on two adjoining sides.

For all items of equipment with an eccentric centre of gravity this symbol shall be marked at the bottom, side and top of the package.

The slinging points shall be marked with a chain symbol.

9.4.2 For dangerous goods

Packages containing goods as per statutory regulations on the packing and transport of dangerous goods are to be marked on all sides with appropriate labels in conformity with IMDG / ADR / RID / IATA regulations.

N.B. : INCORRECT MARKING :

It is expressly understood that the supplier shall be responsible for all expenses caused by improper or wrong marking contents.

10. PACKING LIST

10.1 The packing list will be prepared on contractor's standard forms and will show the summary of the content of each package.

10.2 NUNBERING OF PACKING LIST

- Only one packing list number shall provide by contractor's expediting for each definite batch of goods.
- This P.L.N. "Packing List Number" will be marked on the packages and on the shipping documentation and used as unique number to facilitate the tracking.
- The same number will be shown on the release for packing list issued by contractor's expediting when the batch is technically accepted.

10.3 Two (2) copies of the packing list in waterproof plastic envelope must be securely nailed under a steel sheet on the outer surface of the first package of the batch with the associated detailed item list corresponding to the package, as per indication in ANNEX IV.

10.4 The description of detailed item list of the packing list must correspond exactly to the detailed contents of the package.
The expression "accessories for" or "batch of nuts and bolts" for example, are formally forbidden.

10.5 One copy the detailed item list shall be placed in waterproof plastic envelope inside of each package. Another copy in a waterproof plastic envelope shall be placed outside of the package under a steel sheet.

10.6 Above mentioned shipping documentation shall be in English.

11. PACKING INSPECTION – PACKING PERFORMANCE

11.1 The packing inspection will be confirmed by contractor in due time. It shall take place at supplier's designated premises, as per the following procedure :

- cases and/or crates shall be left open.
- shrink wrapped plastic covering or vacuum plastic covering operations to be in stand by.
- packing lists and packing notes to be handed over to entitled inspector.
  - for marking control.

- for compliance examination of cases contents.

11.2 CONSEQUENTLY, SUPPLIER SHALL :

- After technical acceptance of equipment by contractor, define the expected necessary packing preparation period.
- Prior to scheduled packing inspection date, provide contractor's Expediting department with packing lists and packing notes.
- Advise by fax contractor Expediting department, stating expected date and location of packing acceptance to be performed by contractor's representative.
  - shrink wrapping operation as well as vacuum sealing.
  - closing of cases and/or crates together with incorporation of the appropriate packing lists and packing notes.

12. SHIPPING INSTRUCTION

12.1 ADDITIONAL DEFINITION FOR SHIPPING INSTRUCTION

12.1.1 PURCHASER

A party/company of CONTRACTOR who issue purchase order to VENDOR under the name of Consortium Members

12.1.2 VENDOR

VENDOR means that company(ies) who supplies CARGO under purchase order issued by CONTRACTOR.

12.1.3 FREIGHT FORWARDER

A Company(ies) who is nominated by PURCHASER for Air and Ocean Transportation of CARGO for South Pars Gas Field Development Phase 9 & 10 Project.

12.1.4 SITE

SITE means the area where the plant is located and situated at Bandar Assaluyeh in Iran.

12.1.5 CARGO

Equipment, Materials, Components, Accessories and/or Spare Parts procured by CONTRACTOR

12.1.5.1 GENERAL CARGO

GENERAL CARGO means the CARGO that its weight and dimension is less than 20 metric tons and does not exceed 12m in length, 2.5m in width and 2.5m in height per package.

12.1.5.2 HEAVY and/or OVER DIMMENSION CARGO

HEAVY and/or OVER DIMMENSION CARGO means the CARGO that its weight or dimension is bigger or heavier than GENERAL CARGO.

12.1.6 CONTRACTUAL DELIVERY DATE

The date of delivery which is stipulated in the purchase order

12.1.7 CARGO READINESS

CARGO READINESS means the situation that CARGO can be forwarded to the contractual delivery place agreed between PURCHASER and VENDOR as well as ready for delivery to FREIGHT FORWARDER.

12.2 PURCHASER'S RIGHTS AND OBLIGATION FOR SHIPMENT

12.2.1 Instruction

PURCHASER and/or his FREIGHT FORWARDER reserve the right to issue the adequate instructions for shipment to the VENDOR so that the work will meet PURCHASER's requirement. PURCHASER may, if necessary, coordinate the work between the VENDOR and PURCHASER's FREIGHT FORWARDER and/or Shipping companies. PURCHASER's instructions and coordination shall not relieve the VENDOR of any his obligations under the Purchase Order.

12.2.2 Purchaser's Logistics Organization

PURCHASER establishes his transportation supervisory organization as shown in ATTACHMENT NO.1. The VENDOR shall keep in close contact with PURCHASER's person in charge to perform the transportation work smoothly.

12.2.3 Space Booking

PURCHASER books the space of the Transportation Means for Cargo-Transportation by use of FREIGHT FORWARDER based on "Preliminary Shipping Information for Space Booking", which is as shown in ATTACHMENT NO.5, prepared and provided by the VENDOR.

12.2.4 Notice of Transportation Schedule

After booking the space of Transportation Means, PURCHASER or PURCHASER's FREIGHT FORWARDER shall inform the Vendor of the Transportation Schedule including the name of vessel (or air craft), ETA delivery place (or port) and/or Loading place (or port).

12.2.5 Pre-Shipment Inspection (PSI)

If OWNER or Iranian authorities request PURCHASER to perform PSI, PURCHASER shall coordinate among concerned parties, such as OWNER, VENDOR and Company of Inspection.

12.2.6 PURCHASER will select FREIGHT FORWARDER for the Transportation of the Cargo.

12.3 FREIGHT FORWARDER'S OBLIGATION FOR SHIPMENT

FREIGHT FORWARDER arranges the Transportation Means according to the agreement with PURCHASER, and coordinates the shipment between PURCHASER, carrier and/or VENDOR.

The Scope of FREIGHT FORWARDER is as follows:

- Space booking of the vessel based on the "Preliminary Shipping Information for Space Booking" made by VENDOR. Notify to VENDOR of the arranged vessel and necessary information so that VENDOR may make shipment smoothly.
- Payment of ocean freight to the shipping company and obtaining all originals and some copies of Bill of Lading and dispatch them to PURCHASER.
- Sending a copy of Bill of Lading to VENDOR for VENDOR's collection of payments upon request by VENDOR.
- Getting VENDOR's Certificate of Origin as per designated description for this project on behalf of PURCHASER.
- Getting other "Required Shipping Document", if so requested by PURCHASER.

PURCHASER's FREIGHT FORWARDER's Organization for this project is given in ATTACHMENT NO.2.

12.4 VENDOR'S OBLIGATION FOR SHIPMENT

VENDOR shall follow the FREIGHT FORWARDER's instruction and fully comply with his requests as if the same instructions/requests were made by PURCHASER.

12.4.1 Vendor's Responsible Personnel for Transportation

Within two (2) weeks after receipt of the Purchase Order, the Vendor shall send the information as shown in ATTACHMENT NO.3 to PURCHASER as shown in ATTACHMENT NO.1.

12.4.2 Export License and/or Other Necessary Permits

In case any export licenses and/or other necessary permits, authorization etc. are required by the authorities concerned in the exporting country for the shipment, VENDOR must obtain such licenses, etc. before Cargo Readiness date by VENDOR's own cost and risk.
If any assistance of PURCHASER and/or OWNER for this matter is required, VENDOR shall notify PURCHASER of the details within two (2) weeks after receipt of purchase order and/or

letter of intent issued by PURCHASER.

12.4.3 Transportation Aids

12.4.3.1 Special Traverse

The VENDORS, who supply large quantities of steel plates for tankage or very large equipment such as air cooled heat exchanger or equipment that balancing Center of Gravity is hard and dangerous without Special Traverse for lifting, shall provide the special lifting traverses to be used at both the ports of loading and discharging by Vendor's own cost and risk.

12.4.3.2 Lifting Lugs

The VENDOR who supplies heavy equipment, package type boilers or equipment shall provide lifting lugs on the equipment itself by VENDOR's own cost and risk.

12.4.4 Trade Terms

The trade terms used on the Purchase Order shall be governed and interpreted under the provisions of INCOTERMS 2000.
Besides VENDOR shall note that it is interpreted that the following costs for shipment shall be for account of VENDOR, in case that the trade terms is FOB, unless otherwise agreed between VENDOR and PURCHASER.

- Export Customs Clearance
- Export License, if required
- All Inland Transportation Charges from Vendor's Shops to alongside of Vessel, Container Yard or Airport including Pick Up Charges.
- Container Drayage Charge/Empty Container Drayage Charge.
- Vanning Work/ Vanning Charge for Container Transportation, if any
- CFS (Container Freight Station) Charge for Container Transportation, if any
- All Charges related with Packing and Marking
- Terminal Handling Charge for Container Transportation

In case that FREIGHT FORWARDER pays for the above costs on behalf of VENDOR, those costs shall be compensated by VENDOR to FREIGHT FORWARDER.

12.4.5 Delivery of Cargo

12.4.5.1 Pier Delivery Instructions

After completion of the packing inspection, VENDOR shall contact the FREIGHT FORWARDER for pier delivery instruction. The pier delivery instructions shall not be issued by the FREIGHT FORWARDER unless the FREIGHT FORWARDER receives the shipping documents from the VENDOR. VENDOR shall not release Materials until he receive pier delivery instructions from the FREIGHT FORWARDER.
The FREIGHT FORWARDER shall be authorized to question VENDOR at any stage for the latest delivery information on CARGO on behalf of PURCHASER for smooth shipment.

12.4.5.2 Shipment

- VENDOR shall make shipment on the vessel to be designated by PURCHASER's FREIGHT FORWARDER. The FREIGHT FORWARDER designate the vessel only for shipping arrangement. PURCHASER has nothing to do with any damage and/or losses to the Materials under his custody.

- In case the contract is made on CFR or CIF terms, VENDOR shall ship CARGO at berth terms in loading/discharging port on the conference liner vessel or direct airline which arrives at destination earliest and whose whole derrick can handle the capacity of cargo weight.

- As far as CARGO can be transported by containers, a container vessel will be arranged for shipment. In this case, all relating charges for shipment such as CFS charges, container tax, THC and etc. at the shipping port shall be borne by VENDOR.

12.4.5.3 Partial Transportation (Delivery, Shipment)

Vendor shall deliver CARGO covered by one requisition number in complete set. Partial Transportation (Delivery, Shipment) is basically prohibited unless otherwise specially accepted by PURCHASER. Accessories, special tools and spare parts for erection & commissioning and two (2) years operation spare parts must be shipped together with CARGO.

If partial shipments are allowed by PURCHASER and parts of the order are shipped, VENDOR shall mark as "first partial shipment", "second partial shipment" on the packing list. For the last shipment covering the remainder of the order, VENDOR shall mark "final shipment" on the packing list.

12.4.5.4 Short Shipment

Unless otherwise partial shipment is allowed by PURCHASER, any remaining un-shipped CARGO after VENDOR's final shipment shall be basically delivered on DDP (INCOTERMS 2000) by air. The VENDOR, upon receipt of details, such as duty, warehouse charges, inland transportation charges & etc. paid in importing country, shall remit the same charges to PURCHASER, in case that PURCHASER pay such expenses in advance.

12.4.5.5 Pre-Shipment Inspection (PSI)

VENDOR shall grant inspector(s) for PSI to access to VENDOR's shops or his sub-supplier's shops. Besides VENDOR shall assist Inspector or PURCHASER to perform PSI, and shall follow the inspector's instructions for performance of PSI.

12.4.5.6 Required Shipping Documents

VENDOR shall submit the shipping documents and information specified in the Article 6 hereof.

12.4.5.7 Demurrage

As it is mentioned general term and conditions of purchase, in the event that a delay in loading caused by VENDOR results in retention of the vessel, VENDOR shall be liable for any charges for shipment, if any change in the information including the cargo volume and/or delivery schedule causes extra cost such as the charges for Cancellation, Dead Space, Demurrage and/or Detention, trailer waiting charges, terminal handling charges etc.

12.4.5.8 Storage at the Port of Export

As it is mentioned general terms and conditions for purchasing, should PURCHASER not provide a Transportation Means at the date of shipment, VENDOR shall undertake to store the CARGO for thirty (30) days at VENDOR's expenses and risk from the date of shipment advised by PURCHASER. (Details refer to the general terms and conditions for purchasing)

12.5 REQUIRED SHIPPING DOCUMENTS AND INFORMATION FROM THE VENDOR

12.5.1 Heavy Weight and/or Over-Dimensioned Cargo List and Cargo Sketch

Within sixty (60) days after the issuance of purchase order, the VENDOR shall submit two (2) copies of "Heavy and/or Over-dimensioned Cargo List and Cargo Sketch" giving the data concerning net/gross weight, measurement, center of gravity, position of skids, lifting lugs and etc., to PURCHASER by fax first and Express Courier Service.

12.5.2 Cargo Handling Direction

In case that CARGO requires the special attention as to handling and/or temperature, etc., during the whole transportation and storage, two (2) copies of Cargo Handling Directions shall be submitted to PURCHASER within ninety (90) days prior to the scheduled shipment.

12.5.3 Dangerous and Poisonous Goods

Ninety (90) days before Cargo Readiness date, Vendor shall fill up the necessary information in the ATTACHMENT NO.4 including "IMCO Class" and " UN No." and submit it together with the following documents and information to PURCHASER's Head Office if the Vendor will export dangerous or poisonous goods, defined in IMCO(Inter-Governmental Maritime Consultative Organization) Code :

- ATTACHMENT NO. 4 Safety Data Sheet which is filled up by Vendor
- Contents explanation of chemical and hazardous Cargo
- Detailed explanation on the CARGO including packing style and packing specification
- Catalog showing the name and quality of the CARGO, etc.

12.5.4 Preliminary Shipping Information

Forty five (45) days before Cargo Readiness date, the VENDOR shall send "Preliminary shipping information for Space Booking" as shown in ATTACHMENT NO.5 to PURCHASER Head Office and to the FREIGHT FORWARDER by fax or e-mail.

If only a short time is allowed between the date of the Purchaser Order and the required

delivery date, the VENDOR shall prepare and submit this information within ten (10) days after receipt of Purchaser Order.

The VENDOR's "Preliminary Shipping Information" shall contain at least the following items:

- Expected date of release for shipment (delivery date)
- Material list to be shipped
- Estimated gross weight and dimensions per package. If the gross weight & dimensions cannot be estimated per a package, the estimated total gross weight and measurement per shipment is acceptable.
- Amount of the CARGO to be delivered (if partial shipments are made, the amount shall be declared on a prorated basis)
- Instructions for loading, unloading, handling and special precautions to be observed for storage.

If the notice of any change in cargo volume and/or delivery schedule is given to PURCHASER and/or PAUCHASER's FREIGHT FORWARDER too late for PURCHASER and/or PURCHASER's FREIGHT FORWARDER to correct the necessary arrangement, PURCHASER shall have the right to file a claim against VENDOR for extra cost, such as dead freight or demurrage.

12.5.5 Detailed Packing List and Commercial Invoice (Draft)

Fifteen (15) days before Cargo Readiness date, the VENDOR shall send PURCHASER and PURCHASER's FREIGHT FORWARDER the following documents signed by VENDOR by fax or e-mail first and by courier service. VENDOR shall use the format as shown in ATTACHMENT NO. 6 & ANNEX IV for preparation of these documents.

	To PURCHASER Head Office	FREIGHT FORWARDER
• Commercial Invoice	1 Copy	
• Packing List	1 Copy	1 Copy

12.5.6 Certificate of Origin

Three (3) days before Cargo Readiness date and before the documents are certified by Chamber of Commerce, VENDOR shall prepare and send the drafts of Certificate of Origin to PURCHASER Head Office for PURCHASER's review and/or comment by Fax or e-mail first and by courier service. Vendor can use the format as shown in ATTACHMENT # 7 for preparation of Certificate of Origin.

12.5.7 Dispatch of Shipping Documents

Three (3) days prior to on-board of CARGO, by Fax or E-Mail and by Express Courier Service as follows:

	To PURCHASER Head Office	FREIGHT FORWARDER
• Commercial Invoice (Final)	1 Copy	
• Packing List (Final)	1 Copy	1 Copy
• Certificate of Origin (Final)	1 Copy	

12.5.8 Final Shipping Documents for Payment

VENDOR may present the Final Shipping Documents to the Bank or PURCHASER within the agreed period between PURCHASER and Vendor in accordance with the terms and conditions of Letter of Credit (or Purchase Order).

Besides, within three (3) days after On-Board date, the Vendor shall dispatch the following "Final Shipping Documents" to PURCHASER's head office for pre-review by fax (or e-mail) first and by courier service.
The list of Final Shipping Documents is subject to the terms and conditions of the Project specification issued by the Project Owner.

- Signed Commercial Invoice 1 original and three copies
- Signed detailed and final Packing List 1 original and three copies
- Certificate of Origin certified by Chamber of Commerce in one copy
- IRN (Inspection Release Note) issued by PURCHASER inspector or 3rd party inspector designated PURCHASER in one copy.
- VENDOR's factory inspection certificate in 1 original and three copies
- VENDOR's Guarantee Certificate in 1 original and three copies.

In case that VENDOR does not compensate for the costs which have been paid by FREIGHT FORWARDER on behalf of VENDOR under FOB terms, PURCHASER shall hold payment until the compensation will be made by VENDOR.

12.6 PREPARATION OF TRANSPORTATION DOCUMENTS

12.6.1 Common Respects

12.6.1.1 Consignee, Notify Party and Description of Bill of Lading (or Airway Bill), Commercial Invoice, Packing List, Certificate of Origin, and Others

Consignee Name	:	POGC / NIOC
Address	:	To be informed later
Notify Party Name & Address	:	To be informed later

Stipulate following phrases in all shipping documents :
- NIOC SOUTH PARS PROJECT PHAESE 9 &10
- Final Destination : Pars Special Energy & Economic Zone (Assaluyeh)'

12.6.1.2 Shipper Name of Bill of Lading (or Airway Bill), Commercial Invoice, Packing List, Certificate of Origin, and Others

Shipper Name & Address	:	To be informed later
Address	:	To be informed later

12.6.2 Certification by Chamber of Commerce

Chamber of Commerce which certifies the transportation documents shall be that of the country in which Manufacturer's shop is located unless the separate instruction is given to

VENDOR by PURCHASER.

12.6.3 Bill of Loading

For FOB delivery, Bill of Lading shall be marked with “Freight Prepaid” and Freight will be paid through PURCHASER’s FREIGHT FORWARDER.

12.6.4 Commercial Invoice

Commercial Invoice shall be duly signed by Vendor’s authorized representative with the following affidavit and statement. “We hereby certify that the equipment and/or materials listed herein are of (Country of Origin) origin and/or manufacture and the amounts quoted are true and correct.”

If there is any advance payment for the purchasing items, Commercial Invoice shall show the full amount as well as amount of advance payment already settled and the net amount payable under the invoice.

12.6.5 Packing List

It shall list only those items actually being transported (delivered). Items previously transported or items to be transported in the future shall not be listed.

It shall be prepared by the M/S Excel (Ver 7.0 or equivalent) in attached sample format and sent to PURCHASER by e-mail when VENDOR applies for packing inspection for PURCHASER’s review and check.

In preparing the packing list, the name of CARGO should be same as those described in Purchase Order, even if VENDOR has his own name for CARGO ordered by PURCHASER.

The packing list should include the following information as a minimum.

- Purchase Order and Requisition number
- Consignee name and address
- Destination
- Item or tag number and item name
- Detailed description of every item in each package
- Quantities of each item or subassembly required for complete shipment
- Gross and net weight (KGS) of each package
- Size (CM) / Dimension(Dimension in centimeter) of each package
- Number of package in the shipment
- Origin of item
- Harmonized system code number (HS Number)

12.6.6 Required Documents and Submission Schedule Etc.

VENDOR shall prepare and submit following number of the documents in accordance with designated schedule and method shown as under:

	Required Documents And Information	Submission Schedule	Required Number	Submission Method	Remarks
12.5.1	Heavy weight and/or Over Dimensioned Cargo List and Cargo Sketch (If applicable)	Within 60 days After P.O. date	2 copies	By Courier Service	VENDOR's form is acceptable.
12.5.2	Cargo Handling Direction (If applicable)	By 90 days Prior to the scheduled shipment	2 copy	by FAX	VENDOR's form is acceptable.
12.5.3	Dangerous and Poisonous Goods (If applicable)	By 90 days Before Cargo Readiness date	1 copy	by FAX	VENDOR must fill up necessary Information in the Attachment NO. 4 including "IMCO Class" and "UN No. ".
12.5.4	Preliminary Shipping Information	By 45 days before Cargo Readiness date	1 copy	by FAX	Please refer to Attachment NO. 5.
12.5.5	Detail Packing List and Commercial Invoice (Draft)	By 15 days before Cargo Readiness date	1 original + 3 copies	At first, by fax or e-mail and By Courier Service	Please refer to Attachment NO. 6. & Annex IV
12.5.6	Certificate of Origin	By 3 days before Cargo Readiness date	1 certified original	At first, by faX or e-mail and By Courier Service	
12.5.8	Final Shipping Documents for Payment	By 3 days after On-Board date		At first by Fax or e-mail for pre-review and By Courier Service	

12.7 ATTACHMENTS

The following attachments are parts of this Shipping Instruction.

Attachment NO.1 PURCHASER'S Logistics Organization
NO.2 Freight Forwarder's Organization

NO.3 Notice of Vendor's Responsible Personnel for Transportation
NO.4 Safety Data Sheet
NO.5 Preliminary Shipping Information for Space Booking
NO.6 Commercial Invoice
NO.7 Certificate of Origin

ATTACHMENT NO.1

PURCHASER'S LOGISTICS ORGANIZATION

1. LG Engineering & Construction Corp. (SAMPLE)

PROCUREMENT TEAM

1) Person in Charge : Mr. M. S. Oh / Project Procurement Manager

Address : LG Yeokjun Bldg 537, Namdaemun-Ro 5-Ga, Joong-gu, Seoul, Korea

Telephone : 82-2-728-3733

Facsimile : 82-2-728-1220

E-mail address : msoh@lgenc.co.kr

2) Person in Charge : S. W. Kim / Traffic Coordinator

Address : LG Yeokjun Bldg 537, Namdaemun-Ro 5-Ga,

Joong-gu, Seoul, Korea

Telephone : 82-2-728-3877

Facsimile : 82-2-728-1220

E-mail address : swkim3@lgenc.co.kr

ATTACHMENT NO.2

FREIGHT FORWARDER'S ORGANIZATION

1. KOREA Prime Company for Ex-KOREA CARGO

   Name :
   Address :
   Person in Charge :
   Telephone :
   Facsimile :
   E-Mail Address :

2. EUROPE Agent for Ex-EUROPE CARGO

   Name :
   Address :
   Person in Charge :
   Telephone :
   Facsimile :
   E-Mail Address :

3. JAPAN Agent for Ex-JAPAN CARGO

   Name :
   Address :
   Person in Charge :
   Telephone :
   Facsimile :
   E-Mail Address :

4. SINGAPORE Agent for Ex-SINGAPORE CARGO

Name :
Address :
Person in Charge :
Telephone :
Facsimile :
E-Mail Address :

Remark: Details are will be informed later.

ATTACHMENT NO.3

NOTICE OF VENDOR'S RESPONSIBLE PERSONNEL FOR TRANSPORTATION

TO : LG Engineering & Construction Corp.

Name of Vendor and Agent :

Person in Charge :

Address :

Telephone :

Facsimile :

E-Mail Address :

We are pleased to inform you of the following information at your request in "Transportation Instruction to Vendor"

Purchase Order No.	Description	Quantity

1. Fabricator / Sub Vendor :

Address :

Telephone :

Facsimile :

E-Mail Address :

2. Manager of Transportation :

Person in Charge :

Telephone :

Facsimile :

E-Mail Address :

ATTACHMENT NO. 4

SAMPLE

SAFETY DATA SHEET

Ref. No.
Date

Name of company :

Communication in case of accident Name of person : Tel :

PACKAGE STYLE, ETC.

Job. No.		Req. No.		Item No.	
Contents				Package No.	
Chemical Formula					
Classification by Regulation			UN No.		
IMCO CODE		Fire Service Law			
Net Weight		Net Capacity		Filling Pressure	
Service					

NATURE OR PROPERTY

Appearance		Ignition Temperature	
Odor		Boiling Point	
Specific Gravity		Freezing Point	
Vapor Density		T L V	
Flammable Limit		Oral Toxicity (LD50)	
Flash point			

MAIN HAZARDS

Explosion and Flammability :	
Reactivity :	
Health Hazard :	

EMERGENCY PROCEDURE

Fire	How to Extinguish Extinguishing Agents. Etc.	
Leakage	How to clean Neutralization Agents. etc.	
Protective Clothing	Types of Gas-Mask, etc.	
Acute Toxic	Emergency Measures, Antidote, etc.	

ATTACHMENT NO. 5

SAMPLE

TO : LG Engineering & Construction Corp. Date :
Attn : Mr. M. S. Oh / Project Procurement Manager Ref. No. :
C.C : Mr. S. W. Kim / Traffic Coordinator PAGE :
1 OF 2

Subject : **Preliminary Shipping Information for Space Booking**
SOUTH PARS GAS FIELD DEVELOPMENT PHASE 9 & 10

We hereby send LG E&C the Preliminary Shipping Information for Space Booking as below according to the Transportation Instruction No. : xxxxxx.

P.O Number	:	
Requisition No.	:	
Cargo Description	:	Details as per attached Cargo Details
Total Number of Packages	:	Packages
Total Net Weight	:	KGS
Total Gross Weight	:	KGS
Total Measurement	:	M3
Cargo Readiness Date	:	
Port of Loading	:	

SAMPLE

Cargo Details

Req. No	Item No.	Description	Q'ty	No. of PKG	PKG Style	Net W.T (Kgs)	G.W (Kgs)	Dimension (L x W x H- CM)
7O07-RQ-OO1	SHM-018	2nd REACTOR	1	XXX-S-001 OF 010	WOODEN SKID	92,700	93,300	1828 x 408 x 339
	SHM-019	TOP DAVIT	1	XXX-S-002 OF 010	WOODEN BOX	4,090	4,500	550 x 180 x 150

ATTACHMENT NO.6

COMMERCIAL INVOICE

Shipper/Exporter			No. & date of invoice		
For account & risk of Messers					
Notify party			Remarks		
Port of loading	Final destination				
Carrier	Sailing on or about				
Marks and numbers of PKGS	Description of Goods	Quantity	Unit-price	Amount	

IRAN

N.I.O.C. – PARS OIL & GAS CO.

SOUTH PARS PHASE 9 & 10 PROJECT

LOADING PORT	:	
DISCHARGING PORT	:	
FINAL DESTINATION	:	
P/O NO.	:	(PUECHASER ORDER NUMBER)
DESCRIPTION OF CARGO	:	(GENERAL MATERIAL NAME)
PACKAGE NO.	:	(SERIAL) OF (TOTAL)
GROSS WEIGHT	:	KGS
DIMENSION	:	(L) CM X (W) CM X (H) CM
MADE IN	:	

NIOC SOUTH PARS PROJECT PHASES 9 & 10
Final Destination: Pars Special Energy & Economic Zone (Assaluyeh)

Signed by

VENDOR'S COMPANY NAME MENTIONED IN PURCHASE ORDER

ATTACHMENT NO.7

CERTIFICATE OF ORIGIN

1. Exporter (Name, address, country)	CERTIFICATE OF ORIGIN Issued by XXXXX CHAMBER OF COMMERCE & INDUSTRY Seoul, Republic of Korea
2. Consignee (Name, address, country)	3. Country of Origin
4. Transport details	5. Remarks

6. Marks & numbers; number and kind of packages; description of goods

IRAN

N.I.O.C. – PARS OIL & GAS CO.

SOUTH PARS PHASE 9 & 10 PROJECT

LOADING PORT	:	
DISCHARGING PORT	:	
FINAL DESTINATION	:	
P/O NO.	:	(PUECHASER ORDER NUMBER)
DESCRIPTION OF CARGO	:	(GENERAL MATERIAL NAME)
PACKAGE NO.	:	(SERIAL) OF (TOTAL)
GROSS WEIGHT	:	KGS
DIMENSION	:	(L) CM X (W) CM X (H) CM
MADE IIN	:	

NIOC SOUTH PARS PROJECT PHASES 9 & 10
Final Destination: Pars Special Energy & Economic Zone (Assaluyeh)

7. Quantity

8. Declaration by the Exporter The undersigned, as an authorized signatory, hereby declares that the above-Mentioned goods were produced or manufactured in the country shown in box 3. (Signature) (Name)	9. Certification The undersigned authority hereby certifies that the goods described above originate In the country shown in box 3 to the best of its knowledge and belief. ---------- Authorized Signatory
	Certificate No.

ANNEX I

MAIN SHIPPING MARK

IRAN
N.I.O.C. – PARS OIL & GAS CO.
SOUTH PARS PHASE 9 & 10 PROJECT
LG ENG. & CONSTR. CORP.
CONTRACT NO. POGC-760-81-156

CONTRACT NO.	:	
LOADING / DISCHARGING PORT	:	
FINAL DESTINATION	:	
PACKAGE NO.	:	
P/O NO.	:	
DESCRIPTION OF CARGO	:	
GROSS WEIGHT	:	KGS
DIMENSION	:	(L) X (W) X (H) CM
MADE IN	:	

In case of airfreight shipment the above mentioned specifications remain the same, but instead of point of entry the marking shall be :

AIRPORT OF DESTINATION : TO BE INFORMED LATER

FOR DELIVERY TO : TO BE INFORMED LATER

Two copies of packing lists are to be attached the outside of each packing in waterproof plastic cover underneath a nailed plate. In the case of equipment, which is dispatched unpacked, this plate should be welded on the metal parts (spot welding) or attached permanently to the package in another way depending on the material. One copy of the packing list is to be placed in a clearly visible point in the boxes/crates in a waterproof plastic cover.

ANNEX II
EQUIPMENT STORAGE CODES

	STORAGE IN AIR CONDITIONED WAREHOUSE	STORAGE IN WAREHOUSE	STORAGE OUTDOOR WITH COVERS	STORAGE OUTDOOR WITHOUT COVERS
STRUCTURAL STEEL				
- Panels, beams, columns, grating				X
- Bolding				
- Material for civil work		XXX		X
EQUIPMENT				
- Towers, vessels, kiln, mills, air coolers boilers				X
- Pumps, compressors, blowers, fans, mixers, reducers			XX	
- Tower and vessels internals			XX	
- Bolting and special small items			XX	
- Packing units			XX	
- Mechanical parts		XXX		
PIPING				
- Flanges, fitting 10” and smaller than 10”		XXX		
- Valves 6” and smaller than 6”		XXX		
- Pipe and remaining valves, fittings and flanges				X
- Gaskets and bolting		XXX		
- Welding rods		XXX		
INSTRUEMENTATION / LABORATORY				
- Cable tray and supports			XX	
- Control room panels, racks, cabinets and associated instruments	XXX	XXX		
- All other items		XXX		
ELECTRICAL				
- Cable reels, conduits and cable tray			XX	
- Electrical motors designed for outside service			XX	
- Electrical motors designed for inside service		XXX		
- Supporting steel, stanchion, saddles, etc.				X
- All other materials		XXX		
- Lightning equipment		XXX		
- Panel boards, racks cabinets and associated electrical	XXX		XX	
- Transformers			XX	
- U.P.S., batteries		XXX		
- All other materials				

	STORAGE IN AIR CONDITIONED WAREHOUSE	STORAGE IN WAREHOUSE	STORAGE OUTDOOR WITH COVERS	STORAGE OUTDOOR WITHOUT COVERS
REFRACTORY LINING - Refractory liquid, mortar, castables, mixes, ceramic fibers expended polystyrene, paper adhesive coating, paints… - Dense fire bricks, steel parts, insulating fire bricks and blocks		XXX	XX	
INSULATTIONS - Ceramic fibre and calcium silicate material, metal sheeting - Mineral wool blankets, supporting rings fasteners, banking…		XXX XXX		
PANTING, THINNERS, etc. - All material		XXX		
CIVIL - Anchor bolts - Reinforcing steel - Cement (in bags), bricks		XXX	XX	X
CHEMICAL & CATALYST - Catalysts, alumina balls, molecular sieves		XXX		

ANNEX III
MARKING SPECIFICATION

Mark		Ink		Mark Location	Size	Remark
		Color	Material			
Main Shipping mark		Black	High quality Waterproof Ink or oil paint	Both sides and top	Optimum Size For each Package	
Care Mark	Sling Mark and Center of Gravity mark	Red	High quality Waterproof Ink or Oil paint	Both sides	Optimum Size for each Package	
	Other marks			Top mark : both Sides and ends Others : both Ends		
Approximate weight		Black	High quality Waterproof Ink or Oil paint	Both sides, Ends, and top	15cm and over	Only for packages Exceeding 5 tons: Hand writing is Permissible : unit Shall be of tons
LG E&C Logo		Red, Black	High quality Waterproof Ink or Oil paint	Both sides, Ends, and top	Minimum Diameter 20cm	
Packing list cover		Yellow Letters On black Back-ground	Enamel	Upper part of One end	L = 14cm W = 18 cm	Do not overlap With care mark : Use template

ANNEX IV
PACKING LIST

Shipper/Exporter		No. & date of invoice			
		Remarks :			
For account & risk of Messers					
Notify party					
Port of loading	Final destination				
Carrier	Sailing on or about				
Marks and numbers of PKGS	Description of Goods	Quantity	Net weight	Gross weight	Measurement

IRAN

N.I.O.C. – PARS OIL & GAS CO.

SOUTH PARS PHASE 9 & 10 PROJECT

LOADING PORT	:	
DISCHARGING PORT	:	
FINAL DESTINATION	:	
P/O NO.	:	(PUECHASER ORDER NUMBER)
DESCRIPTION OF CARGO	:	(GENERAL MATERIAL NAME)
PACKAGE NO.	:	(SERIAL) OF (TOTAL)
GROSS WEIGHT	:	KGS
DIMENSION	:	(L) CM X (W) CM X (H) CM
MADE IN	:	

NIOC SOUTH PARS PROJECT PHASES 9 & 10
Final Destination: Pars Special Energy & Economic Zone (Assaluyeh)

Signed by

VENDOR'S COMPANY NAME MENTIONED IN PURCHASE ORDER

PACKING LIST

(Detailed Item List)

No. of **PKGS**	**Description of Goods**	**Quantity**	Net-weight **(KGS)**	Gross-weight **(KGS)**	Size **(CM x CM x CM)**	Measurement **(CBM)**

ANNEX V

PACKING FOR PIPING MATERIAL

A. PIPES (CARBON STEEL) From 1/2" to 4"

1. PACKING IN CRATE – CATEGORY 3 –

with solid panels sides at the extremity of the crate

1.1 Physico-chemical protection :

All parts vulnerable to corrosion must be protected by the manufacturer, who will in particular take responsibility for draining, rinsing, drying, protecting and for other specific and adapted physico-chemical protection.

1.2 Mechanical protection

The pipes will be immobilized by appropriate inserts fixed to the crates.

1.3 Marking

The contractual shipping marking will be fixed to the sides of the crate.

B. PIPES (CARBON STEEL) From 4"to 8"

BUNDLE - CATECORY 2 – (REFER TO SKETCH ANNEX 1)

1. Physico-chemical protection

The pipes shall be provided with initial protection carried out by the manufacturer such as protective oil and internal protection. This protection comprises a spray-applied protective and seales ends (waterproofing, wood, metal or plastic covers) or any chemical protection perfectly adapted to the pipes proposed by the manufacturer approved by Client.

1.1 Mechanical protection

The bundling is carried out with U-IRON or crosswise planks, joined with threaded connecting rods with stop-nuts. Quantities and stamping positions depend on the length, with a 120 cm spacing to prevent distortion.

C. PIPES (CARBON STEEL) ≥ 8"

1. Loose

ENDS TO BE PROTECTED WITH PLASTIC COVERS

1.1 Physico-chemical protection

Refer to above para. 2.1

1.2 Marking

The contractual shipping marking will be directly painted to the pipes if authorized by the contract or label affixed to the bundles.

D. STAINLESS STEEL PIPES (AND OTHER STEELS NOT ALREADY INDICATED) FITTINGS AND FLANGES STAINLESS STEEL AND OTHERS

1. CASE – CATEGORY 4

1.1 Physico-chemical protection

- If needed for this type of pipes, it will be carried out by the manufacturer.

- All parts of fittings and flanges vulnerable to corrosion must be protected by the manufacturer who will in particular take responsibility for draining, rinsing, drying, protection and any adapted protection needed for transportation and storage in safe conditions.

1.2 Mechanical protection

The immobilization will be made by inserts appropriate to the dimensions of the pipes, fittings and flanges.

1.2.1 Austenitic stainless steel pipes

All stainless steel pipes shall be protected against the risks of saline corrosion during cleaning, transport, fabrication, testing and 0 or storage.

1.2.2 Sea-packing shall be used for all pieces of pipes; in case of austenitic stainless steel, the packing must also be water-proof.

ANNEX VI

PACKING FOR STRUCTURAL STEEL

1. PREPARATION OF THE STRUCTURAL STEEL

The technical identification of steelwork pieces will be made according to the instructions provided by the steel structure specification

2. PACKING SELECTION

2.1 Categories for packing

- The structural steel will be packed as per French Industrial Packaging Specification S.E.I 1992 EDITION.
- The 3 following categories for packing will be used.

  - category 1 : bundles) See Sketch N° 2 page 42
  - category 3 : crates) See Sketches N° 4/5 page 42
  - category 4 : cases) See Sketch N° 6 page 43

2.1.1 Physico-chemical protection

When applicable, the structural steel pieces shall be provided with initial chemical protection proposed by the manufacturer, approved by contractor.

2.1.2 Mechanical protection

- BUNDLE – CATEGORY 1

  - Bundles shall be used for packing of standard straight shapes and built up welded shapes not likely to be subject to permanent distortion during handling operations. The packages shall be homogeneous and all parts shall be properly wedged against one another.
  - The bundling is carried out with U-IRON or crosswise planks, joined with threaded connecting rods with stop-nuts. Quantities and stamping positions depend on the lengths,

with a 120 cm spacing to prevent distortion.

- Remark = Placing of accessories between steel shape flanges is prohibited.

◆ CRATE – CATEGORY 3

- With solid panels sides at the extremity of the crate(refer to sketch...)

- Crates shall be used for wrought steel structures or plates, walkways, railing components, cage components, etc., as a rule for all components likely to be subject to distortion during handling.

- The material will be immobilized by appropriate inserts fixed to the crates.

◆ CASES – CATEGORY 4

- 4a / Cases with physico-chemical protection :

  - They shall be used for packing of assembling components, not shop assembled in their final position on main parts, such as short angles, gussets, wedges, etc.,. These components shall previously be grouped in series per identical structure number and coated with the shop protection provided on main parts.

- 4b / Cases with physico-chemical protection and with internal liner (waterproof)

  - They shall be used for bolting materials (screws, nuts, washers) and welding electrodes.

  - Bolting shall be properly oiled and placed in sacks prior to packing. Sacks shall be oiled prior to placing bolts. Only High Strength Friction Bolts shall be kept in their original packing (cardboard boxed).

  - The immobilization will be made by inserts appropriate to the dimensions of the each

concerned type of goods.

- Remark = all parts vulnerable to corrosion must be protected by the manufacturer who will in particular take responsibility for draining, rinsing, drying, protection and any adapted protection needed for transportation and storage in safe conditions.

1. MARKING

The contractual shipping marks will be applied as per para. 9 and ANNEX 1 of the present "PACKING, MARKING and SHIPPING INSTRUCTIONS".